

PayFlow Control

Technical & MVP Delivery

Product: PayFlow Control

Product Type: B2B SaaS MVP Prototype

Positioning: Payment Operations, Risk & Reconciliation SaaS

Industry Focus: Hospitality

Project Type: Independent Product Development Portfolio Project

Delivery Stage: Technical & MVP Delivery

URL PayFlow Control: <https://www.payflowcontrol.my.id/>

URL Repository GitHub: <https://github.com/yusfss02-glitch/payflow-control>

URL YouTrack showcase/public: <https://yusfss02.youtrack.cloud/issues/PC>

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1. Delivery Overview

PayFlow Control was delivered as a functional SaaS MVP prototype based on the approved product strategy, requirements, prioritization, UX direction, and product scope.

The MVP demonstrates how a hospitality business can gain operational visibility across payment transactions and investigate payment-related issues through a single operational interface.

The delivered MVP focuses on five core capabilities:

1. Payment Visibility
2. Reconciliation Management
3. Exception Management
4. Risk Monitoring
5. Compliance Support

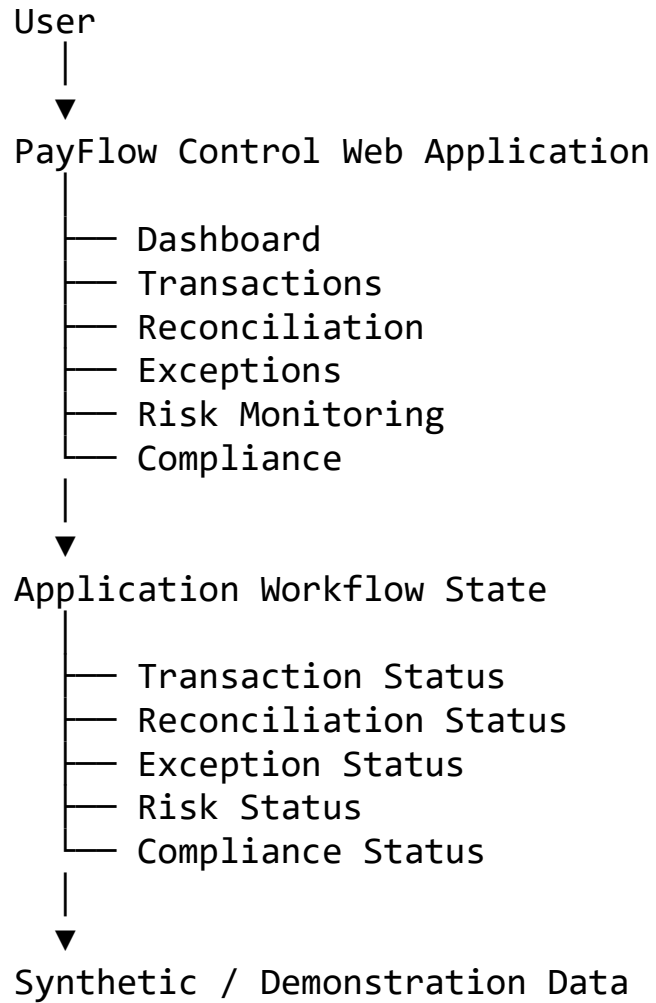
Payment Visibility, Reconciliation Management, and Exception Management represent the primary operational capabilities, while Risk Monitoring and Compliance Support provide supporting decision and control layers.

The MVP is intentionally designed as a **non-production prototype**. It does not process real payments, connect to live banking or payment-provider systems, or use real customer data.

2. Solution Architecture

Current MVP Architecture

The delivered prototype uses a lightweight web application architecture designed to demonstrate product workflows without introducing unnecessary infrastructure complexity.

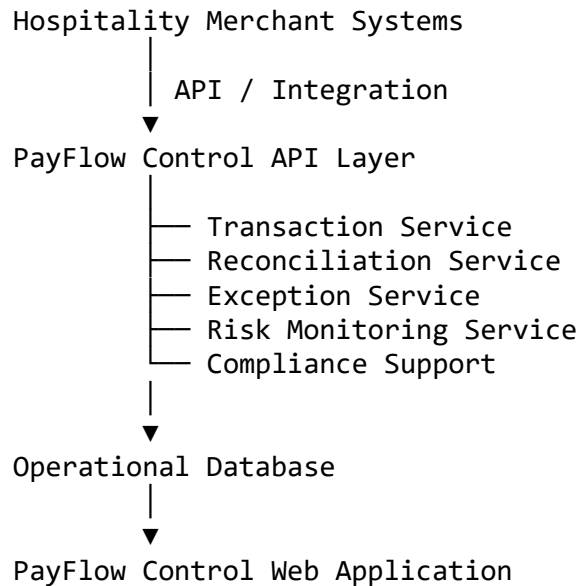


The prototype separates the major operational capabilities into functional application areas while maintaining a consistent user experience and shared workflow state.

The architecture is intentionally lightweight because the objective of this stage is to demonstrate the product operating model and user workflows rather than production infrastructure.

Future Production Architecture

For a production SaaS implementation, the prototype can evolve toward:



This future architecture reflects the intended SaaS operating model: merchant transaction and operational data would enter PayFlow Control through system integrations rather than being manually entered into the dashboard.

3. Technical Architecture Diagram

[SCREENSHOT / DIAGRAM TO INSERT HERE]

Recommended visual:

Create or capture a simple architecture diagram showing:

Hospitality Merchant Systems → API/Integration Layer → PayFlow Control → Transaction / Reconciliation / Exception / Risk / Compliance modules → Database

Clearly label the production components as **Future-State Architecture** and the currently implemented application as **MVP Prototype**.

This distinction prevents the portfolio from implying that the prototype already contains production integrations.

4. Product Data Model

The logical data model was defined around the operational entities required by the product.

Entity	Purpose
Merchant	Represents the hospitality business using the SaaS
User	Represents users accessing the merchant environment
Transaction	Represents an individual payment transaction
Platform / Channel	Represents the external booking/payment channel associated with a transaction
Reconciliation Record	Represents matching between transaction records
Exception	Represents an operational payment issue
Risk Assessment	Represents a payment-risk indicator and assessment
Compliance Record	Represents a compliance-support requirement or review
Subscription Plan	Represents future SaaS entitlement and usage controls

The roadmap explicitly identifies Merchant, User, Transaction, Provider, Risk Assessment, Reconciliation Record, and Subscription Plan as the core entities for the intended SaaS data model.

For the current MVP, these concepts are represented through synthetic application data rather than a production database.

5. Transaction and Integration Model

The MVP uses synthetic transaction data to demonstrate payment operations.

Example:

Transaction ID	Transaction Date	Platform / Channel	Amount	Method	Status
TXN001	April 1, 2026	StayBook	\$1,250	Card	Success
TXN002	April 2, 2026	TripNest	\$980	Bank Transfer	Pending
TXN003	April 3, 2026	RoomLink	\$2,430	Card	Exception

The platform/channel names are synthetic and are used to represent external booking or payment channels.

Future API Integration

In a production environment, the expected integration model would be:

Merchant / PMS / Booking Platform → API → PayFlow Control

Potential API capabilities include:

- Create transaction
- Retrieve transactions
- Retrieve transaction by ID
- Update transaction status
- Submit reconciliation records
- Receive operational events

The original technical roadmap defined POST, GET, GET-by-ID, and PATCH transaction API capabilities as the intended API model.

These APIs are **future production architecture requirements**, not claims that live external APIs are currently connected to the prototype.

6. Security & Access-Control Design

Security was treated as a product requirement and future production capability rather than a production certification claim.

Intended production controls

- Secure authentication

- Session management
- Role-based access control
- Merchant/tenant data isolation
- API authorization
- Audit logging
- Input validation
- Secure secrets and environment configuration

The intended SaaS role model includes:

Role	Primary Responsibility
Merchant Administrator	Manage merchant environment and operational controls
Merchant Analyst	Monitor transactions and investigate operational issues
Platform Administrator	Manage platform-level operations

The roadmap explicitly defines merchant data isolation and role-based access control as SaaS capabilities.

MVP Security Boundary

The current portfolio MVP should **not be presented as production-secure infrastructure**. It is a controlled demonstration environment using synthetic data.

7. Risk & Compliance Data Requirements

Risk and compliance are supporting operational capabilities within PayFlow Control rather than the product's primary purpose.

Risk Monitoring

The MVP demonstrates:

- Risk score
- Risk level
- Risk type
- Transaction association
- Review status
- Resolution workflow

The risk capability uses transparent prototype logic rather than machine-learning prediction or autonomous fraud detection.

Compliance Support

The MVP demonstrates:

- Compliance requirement
- Review status
- Risk classification
- Review workflow
- Approval / rejection states

The feature is positioned as **compliance support and visibility**, not a complete regulatory compliance or AML platform.

8. Development Environment & Tooling

The MVP was developed using an AI-assisted development workflow.

Primary tools

- Next.js
- TypeScript
- React
- Tailwind CSS
- Cursor
- VS Code
- Git
- GitHub
- PowerShell
- Local development environment

Cursor and VS Code were used as the primary AI-assisted development environments. Generated or modified code was reviewed, executed, tested, and iteratively refined rather than being treated as automatically production-ready.

The development workflow follows the principle:

Product Problem → Product Requirements → MVP Capability → Working Prototype
→ Test Evidence → Portfolio Story.

The implementation process was incremental: application foundation, authentication/access concepts, payment visibility, reconciliation, exceptions, risk monitoring, compliance support, followed by integration and MVP review.

9. MVP Implementation Overview

The delivered MVP currently demonstrates the following operational workflows:

Capability	MVP Delivery
Dashboard	KPI overview and operational activity
Payment Visibility	Transaction list, filtering and transaction details
Reconciliation	Matching, review and exception workflow
Exception Management	Exception identification, review and resolution
Risk Monitoring	Risk indicators, scoring and review workflow
Compliance Support	Compliance records and review states
Cross-module Navigation	Linked operational workflows
Workflow State	Status changes reflected across relevant views
Synthetic Data	Safe demonstration dataset

The implementation deliberately prioritizes a credible end-to-end operational experience rather than production-scale infrastructure.

10. Product Screenshots

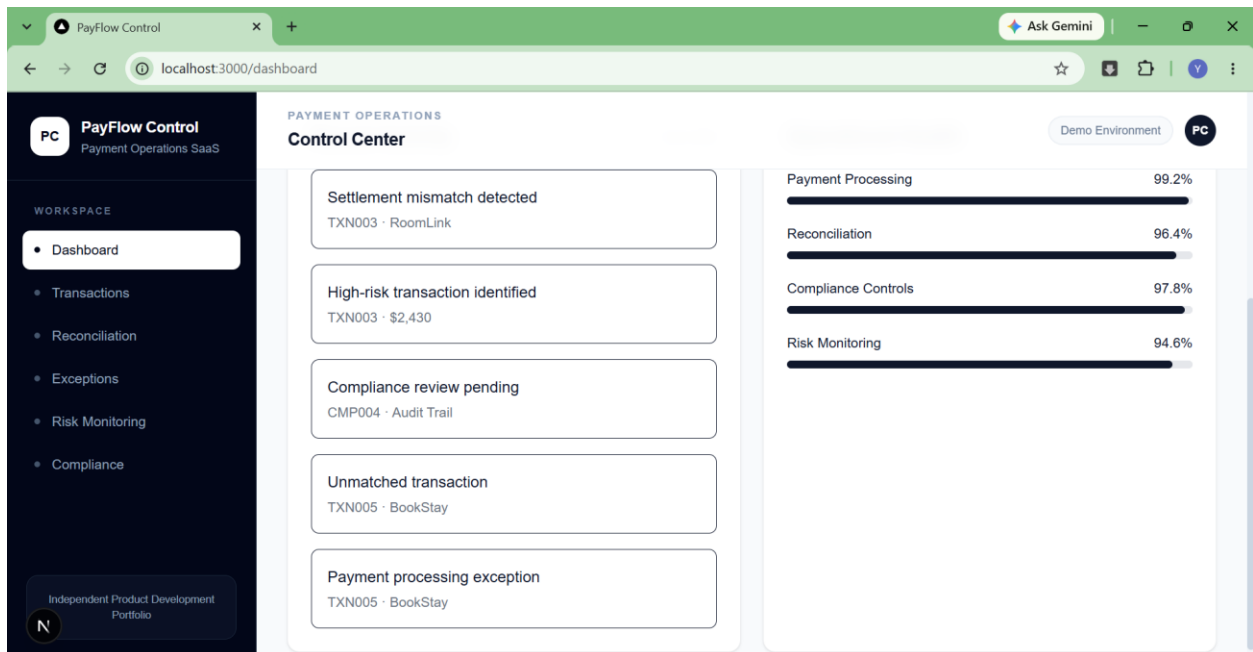
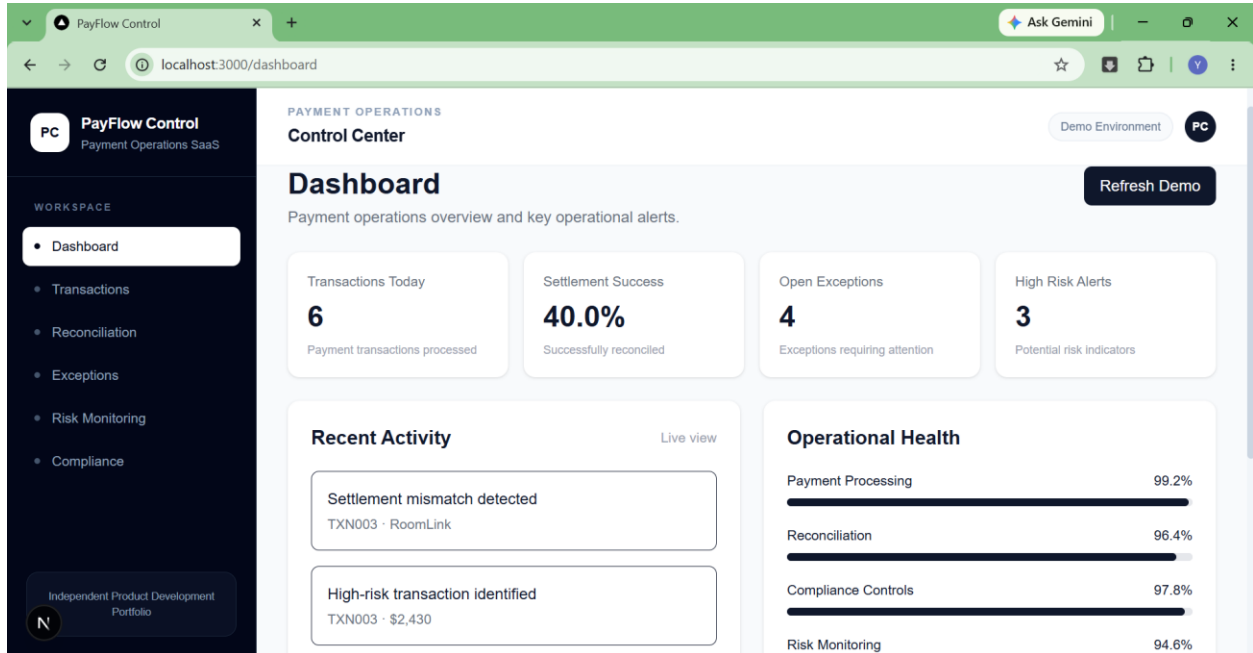
The following screenshots should be captured from the working MVP and inserted into this section.

Screenshot 1 — Dashboard

Capture: PayFlow Control Dashboard showing:

- Transactions Today
- Settlement Success

- Open Exceptions
- High Risk Alerts
- Recent Activity
- Operational Health



Screenshot 2 — Transactions

Capture: Transactions page showing:

- Transaction ID
- Transaction Date
- Platform / Channel
- Amount
- Method
- Status

The screenshot displays the 'Transactions' page in the PayFlow Control application. The page is titled 'Transactions' and includes a subtitle 'Monitor payment transactions across booking platforms and channels.' Below the title, there is a search bar labeled 'Search transaction or platform...' and a dropdown menu for 'All Status'. The main content area features a table with the following data:

Transaction ID	Transaction Date	Platform / Channel	Amount	Method	Status
TXN001	April 1, 2026	StayBook	\$1,250	Card	Success
TXN002	April 2, 2026	TripNest	\$980	Bank Transfer	Pending
TXN003	April 3, 2026	RoomLink	\$2,430	Card	Exception
TXN004	April 4, 2026	TravelHub	\$750	QR Payment	Success
TXN005	April 5, 2026	BookStay	\$1,890	Card	Success

The left sidebar shows the 'Workspace' menu with the following items: Dashboard, Transactions (selected), Reconciliation, Exceptions, Risk Monitoring, and Compliance. The top header shows 'PayFlow Control Payment Operations SaaS' and 'Control Center'.

Screenshot 3 — Transaction Detail

Capture: TXN001 or another transaction detail page showing transaction information and current status.

The screenshot displays the PayFlow Control web application interface. The browser address bar shows the URL `localhost:3000/transactions/TXN001`. The application header includes the "PayFlow Control" logo, "Payment Operations SaaS", and a "Demo Environment" indicator. A left sidebar lists navigation options: Dashboard, Transactions (selected), Reconciliation, Exceptions, Risk Monitoring, and Compliance. The main content area is titled "PAYMENT OPERATIONS Control Center" and features a "Success" status badge. It displays transaction details for TXN001, including the transaction date (April 1, 2026), platform/channel (StayBook), amount (\$1,250), and payment method (Card). A "Transaction Details" section below provides a summary of the transaction ID, date, platform/channel, payment method, amount, and current status (Success).

Transaction Details	
Transaction ID	TXN001
Transaction Date	April 1, 2026
Platform / Channel	StayBook
Payment Method	Card
Amount	\$1,250
Current Status	Success

Screenshot 4 — Reconciliation

Capture: Reconciliation Queue showing records with:

- Matched
- Unmatched
- Under Review

The screenshot shows the PayFlow Control web application interface. The browser address bar indicates the URL is localhost:3000/reconciliation. The application header includes the PayFlow Control logo, the text 'Payment Operations SaaS', and a 'Demo Environment' badge. A sidebar on the left lists navigation options: Dashboard, Transactions, Reconciliation (selected), Exceptions, Risk Monitoring, and Compliance. The main content area is titled 'Reconciliation' and includes a subtitle 'Monitor payment matching and reconciliation exceptions.' Below this, three summary cards display counts: Matched (2), Unmatched (3), and Under Review (0). A 'Reconciliation Queue' table follows, with a status filter set to 'All Status'. The table contains three records: REC001 (Matched, \$1,250), REC002 (Matched, \$980), and REC003 (Unmatched, \$2,430). Each record has an 'Action' button labeled 'Matched' or 'Review'.

Record	Transaction	Platform / Channel	Amount	Status	Action
REC001	TXN001	StayBook	\$1,250	Matched	Matched
REC002	TXN002	TripNest	\$980	Matched	Matched
REC003	TXN003	RoomLink	\$2,430	Unmatched	Review

Screenshot 5 — Exception Management

Capture: Exception list and/or an exception detail page showing an operational issue and its workflow status.

The screenshot displays the PayFlow Control web application interface. The browser address bar shows 'localhost:3000/exceptions'. The application has a dark sidebar on the left with a 'PayFlow Control' logo and a 'Workspace' menu containing 'Dashboard', 'Transactions', 'Reconciliation', 'Exceptions' (selected), 'Risk Monitoring', and 'Compliance'. The main content area is titled 'PAYMENT OPERATIONS Control Center' and 'Exceptions', with a subtitle 'Monitor and resolve payment operation exceptions.' Below this is an 'Exception Queue' section with a status filter set to 'All Status'. The queue contains four entries, each with an 'Exception ID', 'Transaction', 'Platform / Channel', 'Type', 'Priority', 'Status', and an 'Action' button labeled 'Start Review'.

Exception ID	Transaction	Platform / Channel	Type	Priority	Status	Action
EXC001	TXN003	RoomLink	Settlement Failure	High	Open	Start Review
EXC002	TXN006	InnFinder	Payment Exception	Medium	Open	Start Review
EXC003	TXN002	TripNest	Reconciliation Difference	Medium	Open	Start Review
EXC004	TXN005	BookStay	Payment Exception	Medium	Open	Start Review

Screenshot 6 — Risk Monitoring

Capture: Risk Alerts page showing:

- Risk ID
- Transaction
- Risk Type
- Score
- Status

PayFlow Control
Payment Operations SaaS

WORKSPACE

- Dashboard
- Transactions
- Reconciliation
- Exceptions
- **Risk Monitoring**
- Compliance

Independent Product Development Portfolio

PAYMENT OPERATIONS
Control Center

Demo Environment

Risk Monitoring

Monitor payment risk signals and investigate high-risk transactions.

Open: **3** Under Review: **0** Resolved: **0**

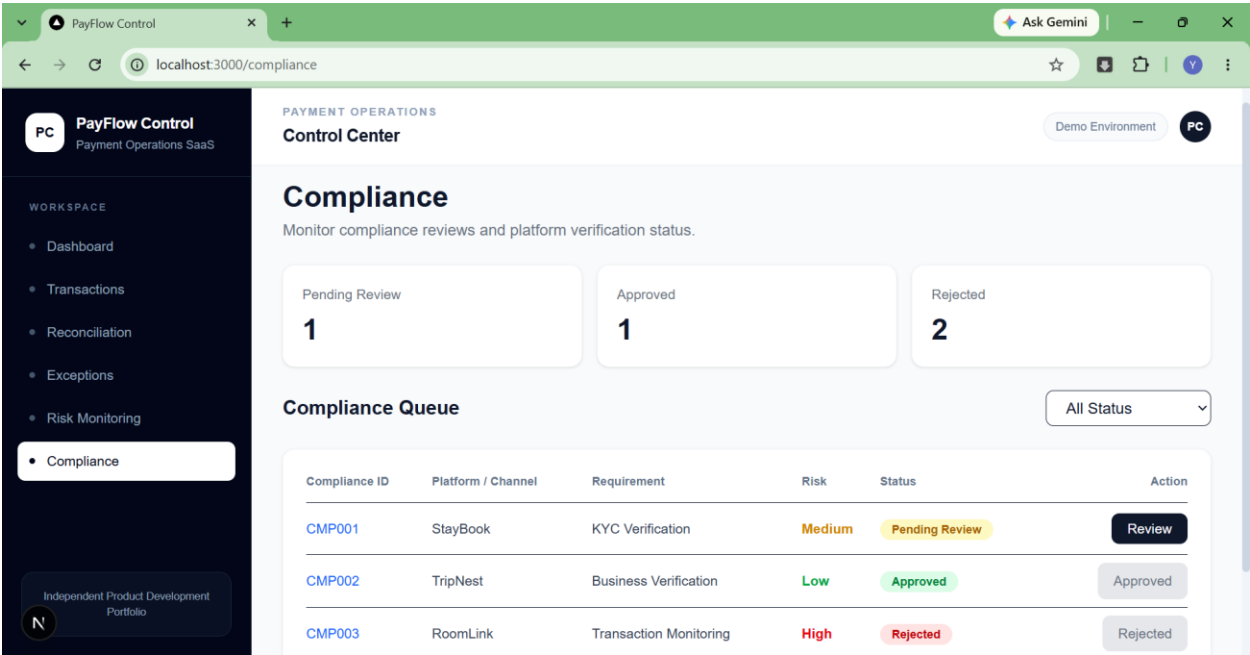
Risk Alerts All Status

Risk ID	Transaction	Platform / Channel	Risk Type	Score	Status	Action
RISK001	TXN003	RoomLink	High Value Transaction	92	Open	<button>Review</button>
RISK002	TXN006	InnFinder	Unusual Payment Pattern	84	Open	<button>Review</button>
RISK003	TXN002	TripNest	Velocity Alert	67	Open	<button>Review</button>

Screenshot 7 — Compliance

Capture: Compliance Queue showing:

- Compliance ID
- Platform / Channel or relevant entity
- Requirement
- Risk
- Status



11. Key Technical Decisions

Decision	Rationale
Lightweight web architecture for MVP	Demonstrates product workflows without unnecessary infrastructure
Synthetic data	Avoids exposure of real customer/payment information
Platform/channel naming	Better represents the hospitality payment ecosystem
Shared workflow state	Allows operational actions to affect relevant dashboard views
Transparent risk rules	Makes prototype decisions explainable

Decision	Rationale
Modular product areas	Allows payment, reconciliation, exception, risk and compliance capabilities to evolve independently
AI-assisted development	Accelerates prototype delivery while retaining human review and testing
Future API-first integration	Supports integration with merchant/PMS/booking systems in a production architecture

12. Technical Risk Register

Risk	Impact	MVP Mitigation	Future Action
Synthetic data does not represent all production scenarios	Medium	Clearly label synthetic scenarios	Validate against real operational requirements
No live payment-provider integration	High	Use simulated transaction workflows	Develop secure API integrations
Prototype state is not production persistence	High	Keep MVP scope controlled	Introduce production database
Authentication/security is not production-grade	High	Use controlled demo environment	Implement production authentication and authorization
Risk logic is rule-based	Medium	Keep rules transparent	Validate and improve risk methodology
No production-scale infrastructure	Medium	Focus on functional MVP	Conduct scalability and reliability engineering
Compliance capability is limited	High	Position as compliance support	Conduct regulatory and legal requirements analysis

13. MVP Limitations

The current MVP is intentionally not production-ready.

It does not include:

- Real payment processing
- Live banking connections
- Live payment-provider connections
- Real merchant/customer data
- Production-grade scalability
- Production security certification
- Full fraud-management capability
- Full regulatory compliance automation
- Enterprise-scale workflow orchestration
- Production disaster recovery
- High-availability infrastructure

These boundaries are deliberate MVP scope decisions rather than missing product requirements.

14. Deployment

The PayFlow Control MVP was deployed to Vercel using the existing Next.js application and GitHub-based deployment workflow. Vercel was selected because it provides native support for Next.js, streamlined deployment from GitHub, automatic production builds, and managed HTTPS/SSL without requiring separate application hosting infrastructure.

From a strategic perspective, Vercel keeps the portfolio MVP lightweight and cost-efficient while providing a professional, browser-accessible production demonstration. This allows the project to focus resources on product capability and delivery evidence rather than unnecessary infrastructure management. The deployment remains a portfolio demonstration environment and should not be interpreted as production-grade payment infrastructure.

Live SaaS: <https://www.payflowcontrol.my.id/>

15. Delivery Outcome

The Technical & MVP Delivery stage produced a working SaaS prototype that translates the approved product strategy into a demonstrable operational product experience.

The delivery demonstrates the complete product-development progression:

Product Problem → Product Requirements → MVP Capability → Working Prototype → Test Evidence → Portfolio Story.

The result is a functional portfolio MVP demonstrating how PayFlow Control could evolve into a hospitality-focused payment operations platform through API-based integrations, stronger data architecture, production security, and scalable SaaS infrastructure.

16. Recommended Production Evolution

If PayFlow Control were advanced beyond the portfolio MVP, the next technical evolution would focus on:

1. Production database and persistent data architecture
2. Authentication and role-based access control
3. Merchant/tenant isolation
4. API integration framework
5. Hospitality PMS and booking-platform integrations
6. Provider/payment-channel integrations
7. Production-grade observability and audit logging
8. Security hardening
9. Automated testing and CI/CD
10. Scalability, reliability and deployment architecture

These items represent the transition from a **demonstrable product-development MVP** to a **production SaaS platform**.

17. AI Assistant

PayFlow Control is also equipped with an **AI Assistant** that enables users to ask questions directly within the product and receive instant answers without having to manually search or open the product documentation. The assistant is designed to provide a more accessible and efficient way for users to explore product capabilities, features, workflows, and other documented information.

For this portfolio prototype, the AI Assistant is intentionally implemented as a **simulation of an embedded product AI capability** and is constrained to the information available in the PayFlow Control product documentation. This ensures that responses remain aligned with the documented product scope and do not introduce unsupported product facts.

The AI Assistant is powered by **Google Gemini 3.1 Flash Lite**. In a production SaaS environment, the capability could be further developed according to user and business requirements, including broader knowledge sources, more advanced contextual understanding, workflow assistance, and integration with operational data or other product capabilities.

